Temperature Dependence of Doppler-Broadening for Rubidium-85 Absorption

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Introduction

The absorption of rubidium-85 atoms by a diode laser centered at 780nm was measured while sweeping through frequencies, exciting the atoms from the ground state to the first excited state. It is observed that due to random thermal motions of rubidium atoms, an absorption profile takes a Gaussian shape. As the temperature of the rubidium atoms increases, the frequency spread of the absorbed laser light is increased due to the Doppler Effect. The width of the Gaussian profile is proportional to $\sqrt{T}$.[1]

Theory

The Maxwell Velocity Distribution (the number of molecules per unit volume with velocity components between v_x and $v_x + dv_x$) for a gas in equilibrium is represented by

$$f(\mathrm{v}_x)\mathrm{d}\mathrm{v}_x = n\left(\frac{m}{2\pi kT}\right)^{3/2} e^{-\left(\frac{\frac{1}{2}mv_x^2}{kT}\right)} dv_x,$$

where *k* represents the Boltzmann Constant, *T* represents the temperature of the rubidium, *v* is the velocity of the rubidium atoms, and *m* is the mass of the rubidium atoms.[2]
Through the Doppler Effect, the velocity of the atoms can be related to the frequency of laser light needed to excite them by

$$v = v_0\left(1 - \frac{\mathrm{v}_x}{c}\right) \text{ or } \mathrm{v}_x = c\left(1 - \frac{v}{v_0}\right) \text{ and } |d\mathrm{v}_x| = \frac{c}{v_0}dv.$$

The distribution of the observed Gaussian Doppler well is represented by

$$f(v)dv = n\left(\frac{m}{2\pi kT}\right)^{3/2} \frac{c}{v_0} e^{-\left(\frac{\frac{1}{2}mc^2\left(1-\frac{v}{v_0}\right)^2}{kT}\right)} dv.$$

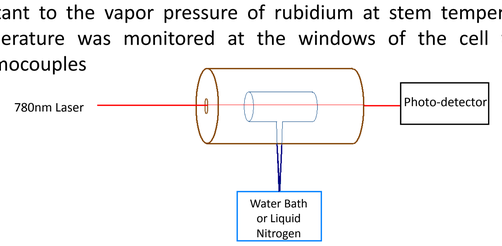
The full-width at half-maximum (Δf_{FWHM}) of the Doppler well is

$$\Delta f_{FWHM} = (2\sqrt{ln2})f_o\sqrt{\frac{2kT}{mc^2}} = 2f_o\sqrt{\frac{2ln2kT}{mc^2}}.$$

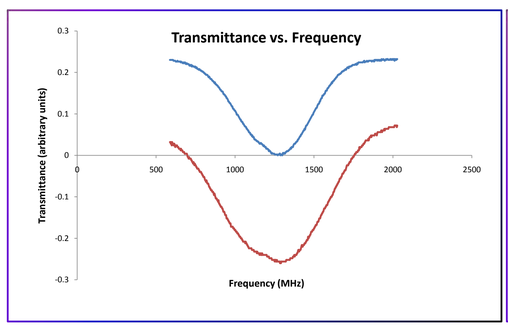
When observing Doppler-broadening, a relation can be made between FWHM and $\sqrt{T}$.

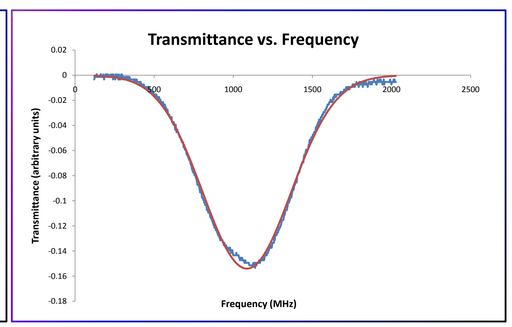
Experiment

A diode laser at 780 nm was used to excite the $5S_{1/2}$ to $5P_{3/2}$ transition of rubidium-85. Saturated absorption in a separate rubidium cell was used to verify the wavelength of the laser. The rubidium cell in which temperature dependence of Doppler-broadening was observed consisted of a Pyrex cylinder with a long arm that extended perpendicularly to the length of the cell. The cylindrical component of the cell was encased within an oven made of Electrotemp Cement by Sauereisen Cement. The temperature of the cell was increased by wrapping a Thermolyne BriskHeat flexible electric heating tape around the outside of the oven. The stem of the cell was cooled by two different methods. First, the stem was kept in a beaker of ice water. Second, the stem was extended into a cryostat with glass beads supplied with liquid nitrogen. By keeping the temperature of the stem a constant, the pressure of the cell was held constant to the vapor pressure of rubidium at stem temperature. The temperature was monitored at the windows of the cell with k-type thermocouples



The laser was allowed through both the oven and rubidium cell and into a photo-detector, which gathered data of the rubidium's absorption. A line was modeled to be the sum of three Gaussians, because the large Doppler well is actually composed of three Doppler wells due to hyperfine splitting.

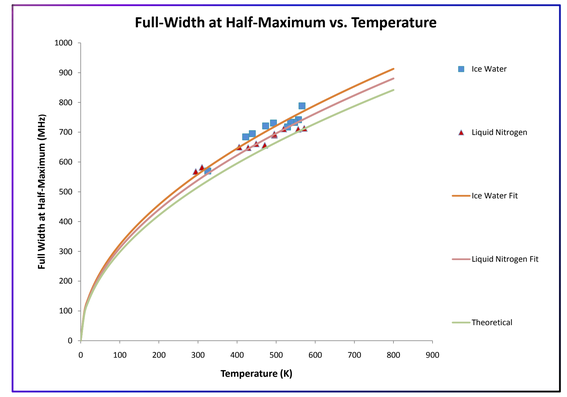




The graph on the left shows the data for 311K (blue line) and 520K (red line). This demonstrates the wider well as temperature increases. The graph on the right shows transmittance against frequency at 406 K. The blue line shows the data gathered, and the red line shows the fit of the sum of three Gaussian wells.

Results

The graph below displays the dependence of Doppler-broadening on the square root of temperature, showing the data gathered in the experiment as well as a line of best fit. Using the ice water bath, the proportionality constant was found to be 8.4% higher than the theoretical value. Using the liquid nitrogen, the proportionality constant was found to be 4.6% higher than the theoretical value.



Conclusions

Our work has shown that:
•Our experimental data follows a line of best fit of $\sqrt{T}$.
•The values for the coefficient of $\sqrt{T}$ for the stem resting in the ice water bath and the liquid nitrogen are off by 8.4% and 4.6%, respectively. The liquid nitrogen allowed the rubidium in the cell to have a lower vapor pressure. This resulted in fewer atoms in the heated part of the cell, allowing for better transmission.
Possible pathways for this project could be to:
•Find a more efficient way to heat the cell and get an accurate reading of temperature.
•Find a method of cooling the cell below room temperature to determine the agreement of the fit for temperatures less than what has been gathered so far.

References

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[2]Krane, K. 1983. Modern Physics. John Wiley & Sons. New York, New York. pgs. 336-341

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